& Crutchfield, 2017), which provide a tractable yet rich setting for studying sequence prediction. As shown in Fig. 1A, these HMMs consist of three hidden states with observable emissions controlled by parameter α and transitions by parameter x. Higher values of $\alpha \in [0, 1]$ mean each state more strongly prefers its unique emission symbol, providing clearer information about the generating state. The parameter $x \in (0, \frac{1}{2}]$ controls state persistence—low values create high inertia where states tend to persist, while high values increase transition probabilities between states. For each experimental run, we generate sequences by sampling from an HMM with specific (α, x) values.

Training Process. We train single-layer transformers on next-token prediction using gradient descent, with sequences sampled from our parametrized HMMs as training data. The model learns to predict the next token in each sequence by minimizing cross-entropy loss (see App. B for details). Our primary analysis focuses on single-layer architectures to clearly isolate how the attention mechanism implements constrained belief updating, though we also validate that these mechanisms persist as the foundational computation in deeper networks (Figs. A5;A6).

Analysis of Representations and Computations. To study how the model processes information, we analyze intermediate and final activations in the residual stream (Fig.1C). We apply principal component analysis (PCA) to these activations across all possible input sequences, finding that the representations are well-captured by a low-dimensional space. In some cases, we slightly rotate the PCA basis to align with theoretically meaningful directions. This dimensionality reduction enables us to visualize how representations evolve through the network—from input embeddings, through the intermediate state after attention, to the final output state after the MLP layer (Fig.1D). To understand how the network manipulates these representations, we analyze the learned weights and attention patterns, examining how attention transforms input embeddings into intermediate representations and how the MLP layer transforms these into the final geometry. At each stage, we compare the learned representations to theoretical predictions derived from optimal Bayesian updates[1].

4. Results

4.1. Intermediate representations are fractals, but not belief state geometry

Through PCA of the residual stream, we observe two distinct fractal structures in transformers trained on Mess3 HMM data: one after the attention mechanism but before the MLP, and another in the final layer output (Figs. 1, 4). While the final representations align with the geometry of theoretical belief states, the intermediate fractals exhibit a markedly different structure. The systematic difference between intermediate and final representations raises two key questions: (1) How does attention construct these intermediate fractals and (2) why do they take these particular geometric forms? The following results reveal the algorithmic process behind their construction and provides a theoretical explanation for their previously unexpected structure.

4.2. Intermediate representations are built by algorithms in the belief simplex

To determine how the intermediate representation is constructed, we performed mechanistic interpretability on the attention heads. We find that attention performs an algorithm with a direct interpretation in the belief simplex.

At every context position, the residual stream can be thought of as a d_{model}-dimensional skip connection communication channel streaming alongside all layers, carrying all working memory in a transformer (Elhage et al., 2021). Attention and MLP modules read in linear transformations of the residual stream and then add their output to the local residual stream at each layer (Vaswani et al., 2017).

Following (Elhage et al., 2021), we decompose the attention operation into two circuits: (i) the output-value (OV) circuit, which specifies what information is read from each position and how it transforms into a vector that can be broadcast to other positions, and (ii) the query-key (QK) circuit, which computes the similarity of a linearly transformed source and destination to determine how much to update the destination's residual stream with that source's OV contribution.

For an attention head, the residual stream update $\vec{x}_{\text{d}}^{(\text{mid})} = \vec{x}_{\text{d}}^{(\text{pre})} + \vec{c}_{\text{d}} \in \mathbb{R}^{d_{\text{model}}}$ at the *destination* position s is:

$$\vec{c}_{\text{d}} = \sum_{\text{s} \leq \text{d}} A_{\text{d,s}} \vec{v}_{\text{s}} \tag{3}$$

Here, $\vec{v}_{\text{s}} = W_{\text{O}} W_{\text{V}} \vec{x}_{\text{s}}^{(\text{pre})}$ represents the OV circuit's contribution from *source* position s, where W_{O} and W_{V} are the attention output and value weight matrices respectively, and $\vec{x}_{\text{s}}^{(\text{pre})}$ is the incoming residual stream vector at position s. Attention $A_{\text{d,s}}$ is determined by the QK circuit through query–key inner product and the causally masked softmax operations:

$$A_{\text{d,s}} = \delta_{s \leq d} \frac{e^{\vec{q}_{\text{d}} \cdot \vec{k}_{\text{s}} / \sqrt{d_{\text{h}}}}}{\sum_{s'=1}^{\text{d}} e^{\vec{q}_{\text{d}} \cdot \vec{k}_{s'} / \sqrt{d_{\text{h}}}}} , \tag{4}$$

where $\vec{q}_{\text{d}} = W_{\text{Q}} \vec{x}_{\text{d}}^{(\text{pre})}$ is the query vector from position d, $\vec{k}_{\text{s}} = W_{\text{K}} \vec{x}_{\text{s}}^{(\text{pre})}$ is the key vector from position s, d_{h} is the head dimension, and W_{Q} and W_{K} are each $d_{\text{h}} \times d_{\text{model}}$ weight matrices. Recall that attention is non-negative $0 \leq A_{\text{d,s}} \leq 1$

[1] Code for analysis of can be found here.

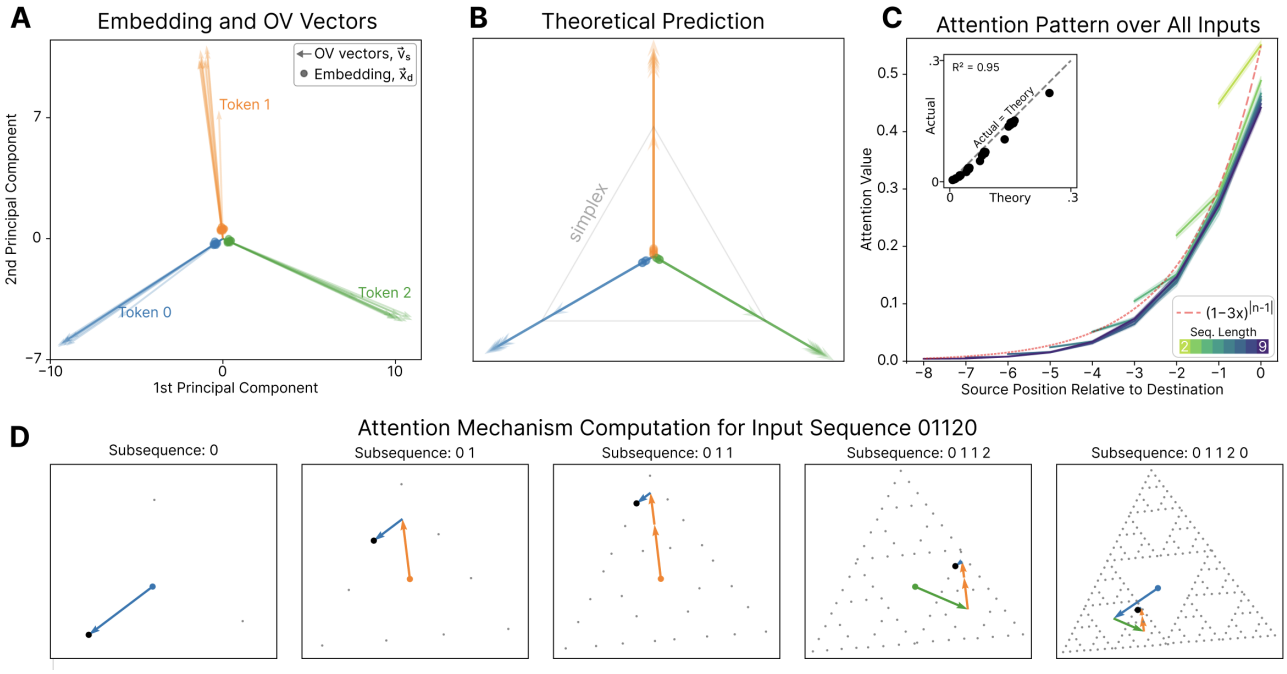

Figure 2: **Intermediate Representation Construction by Attention.** A transformer trained on Mess3 with $x = 0.15$ and $\alpha = 0.6$ exhibits intermediate representations constructed through a specific attention mechanism. **(A)** The OV vectors (arrows) form three distinct clusters, each corresponding to a token and positioned at the vertices of a triangle, while token embeddings (circles) are clustered near the origin. **(B)** Our theoretical predictions for the OV vectors (shown for all (position, token) pairs) and embeddings (for positions > 2) align closely to those found in the trained transformer. **(C)** Attention patterns are primarily determined by the positional distance between the destination and source tokens, following an exponential decay described by $(1 - 3x)^{|n-1|}$. They are largely independent of specific token sequences. **(C, inset)** The theoretical (Eq. (13)) and actual values in the attention pattern align closely. **(D)** Construction of intermediate representations for five input subsequences of increasing length (from the example sequence 01120, shown left to right). The attention mechanism builds the fractal by taking linear combinations of the three $\vec{v}_s$ vectors. The colored vectors illustrate the components of the sum for each example subsequence, while the gray dots represent all possible vector sums for all sequences at that position.

and, for each destination position d, the attention to all sources sums to one: $\sum_{s \leq d} A_{d,s} = 1$. Eq. (3) shows how each attention head computes its update by weighting the transformed values ($\vec{v}_s$) from all previous positions according to their relevance ($A_{d,s}$) to the current position.

Our analysis yields several key insights into how the attention mechanism constructs the intermediate representations. First, we find that projecting token embeddings (the inputs into the attention head) onto PCA space reveals three clusters that lie close to the origin, as shown in Fig.2A. Meanwhile, the OV projections form update vectors $\vec{v}_s$ that cluster in three directions pointing toward the vertices of a triangle, naturally interpreted as the vertices of the belief simplex in Fig. 2. The model combines these directions through weights $A_{d,s}$ determined by the QK circuit as described by Eq. (3). For Mess3, these attention weights are invariant to token identity and decay exponentially with distance from the current position, controlling how past information is integrated. As the attention weight decays with distance, the impact of past tokens on the current belief state diminishes over time. Through this process of weighted vector addition within the belief simplex, the attention mechanism constructs the intermediate representations, resulting in the observed fractal structure shown in Fig. 2D. Incredibly, **the computation the attention head performs is completely interpretable as a dynamic process in the belief simplex**.

4.3. Relating Intermediate Representations to Belief Updating Equations

The interpretation of attention as operating in the belief simplex suggests a connection to the theory of belief updating. Since the OV circuit is only able to access information from the source token that is attended to, we can write a constrained belief updating equation that sums contributions from the value of the token $n = d - s$ places back for each value of n, assuming the initial belief is the stationary distri-

bution of the HMM, $\boldsymbol{\pi}$. This gives the following equation for the constrained belief at position d in the sequence:

$$\vec{r}_1^{(z_{1:\mathrm{d}})} = \boldsymbol{\pi} + \sum_{\mathrm{s}=1}^{\mathrm{d}} \left(\boldsymbol{\pi} T^{|z_{\mathrm{s}}} T^{\mathrm{d}-\mathrm{s}} - \boldsymbol{\pi}\right) \tag{5}$$

where T is the HMM's hidden state transition matrix (marginalizing out the emissions), and $T^{|z}$ is the HMM transition matrix conditioned on seeing token z (see App. A for details). Eq. (5), interpreted as a context-induced point in a vector space, is the natural geometric embedding of

$$\Pr(S_{\mathrm{d}}) + \sum_{\mathrm{s}=1}^{\mathrm{d}} \left[\Pr(S_{\mathrm{d}}|Z_{\mathrm{s}}{=}z_{\mathrm{s}}) - \Pr(S_{\mathrm{d}})\right] . \tag{6}$$

This equation describes the best possible embedding if you haven't seen any context, $\Pr(S_{\mathrm{d}}) = \boldsymbol{\pi}$, followed by independent corrections to that prediction from the token at each preceding context position, $\Pr(S_{\mathrm{d}}|Z_{\mathrm{s}} = z_{\mathrm{s}}) - \Pr(S_{\mathrm{d}}) = \boldsymbol{\pi} T^{|z_{\mathrm{s}}} T^{\mathrm{d}-\mathrm{s}} - \boldsymbol{\pi}$. Notably, since Eq. (6) is a distribution over latent states S_{d} rather than merely the next token $Z_{\mathrm{d}+1}$, this constrained-update equation naturally implemented by attention implies a probability density over all extended futures $Z_{\mathrm{d}+1:\infty}$ rather than just the next timestep.

It is useful to take a step back and get a handle on the intuition for Eq. (5) and Eq. (6). Bayesian inference (Eq. (2)) requires multiplying token-specific transition matrices, a fundamentally recursive process where updates depend on the full history integrated up to the previous step. In contrast, an attention head computes its output at position d via a parallel, feedforward weighted sum (Eq. 3) of value vectors, v_s from source positions $s \leq d$. Crucially, each v_s contains only local information from position s. It cannot directly access or depend on the specific tokens between s and d due to the parallel nature of the value computation and attention weighting. Therefore, the most information that token z_s can independently contribute to the belief at position d within this single-layer constraint is the correction derived from knowing z_s occurred $d - s$ steps prior, assuming a default starting belief π and no knowledge of intervening tokens. These independent displacements from the stationary distribution over latent states is the difference of probability distributions: $Pr(S_d|Z_s) - Pr(S_d)$. Linear algebraically, this contribution is precisely $\pi T^{|z_s} T^{d-s} - \pi$. Summing these over all past sources naturally yields the constrained belief update form in Eq. (5) (see Fig. A1 for a conceptual diagram). It represents the best possible parallel approximation achievable by a single attention layer given its architectural limitations.

Eq. (5)'s constrained belief geometry closely matches the intermediate structure observed in the central range of $\alpha \in [0.2, 0.6]$, as shown in the left two columns in Fig. 4. As α moves further from this range, we observe gradually increasing deviations between predicted and actual representations, though the overall structure remains similar. A complete characterization of how these deviations scale with α remains for future work.

Fig. A3 demonstrates that transformers reliably discover these theoretical geometries during training, with MSE to the constrained belief theory decreasing rapidly for post-attention activations while MSE to the full belief geometry simultaneously decreases for post-MLP activations. The final converged representations show strong quantitative agreement with our theoretical predictions (Fig. A4), with MSE values orders of magnitude lower than random initialization for both the constrained and full belief geometries.

4.4. Attention Implements a Spectral Algorithm to Build the Constrained Beliefs

Eq. (5) shows how the attention pattern in our model must relate to powers of the Markov transition matrix of the underlying hidden states, T^n, where n is the relative token distance.

To understand how this works, we turn to spectral analysis. The main goal is to understand how information or influence from a past token (at source position s) propagates to affect the belief state at the current destination position d. This influence mathematically depends on the sequence of hidden state transitions between s and d, captured by powers of the HMM transition matrix, T^{d-s}.

Spectral decomposition (using eigenvalues λ and associated projectors T_λ) is a standard mathematical tool to analyze matrix powers because it simplifies T^n into a sum $\sum_\lambda \lambda^n T_\lambda$. The eigenvalues ($\lambda$) are crucial because they tell us the rate at which the influence of past information decays (if $|\lambda| < 1$) or even oscillates (if λ is negative or complex) as the distance $n = d - s$ increases.

Our key finding is that the learned attention weights ($A_{d,s}$ in Eq. 3) directly implement this propagation effect, effectively learning to approximate the λ^{d-s} decay predicted by the theory. This explains why attention patterns often show exponential decay, and why multiple heads are needed to capture oscillatory patterns arising from negative eigenvalues. Furthermore, this spectral perspective allows us to make precise, verifiable predictions about the learned OV vectors and token embeddings, directly connecting the dynamics of the data (via T's eigenvalues) to the specific parameters learned by the transformer.

When T is diagonalizable with a set of eigenvalues Λ_T, it then has a simple spectral decomposition such that we can